Part: **3.2 – MECHANICALLY FASTENED JOINTS**

Chapter: **3211**

Title: **RIVETS LOADS IN CONTINUOUS JOINTS**

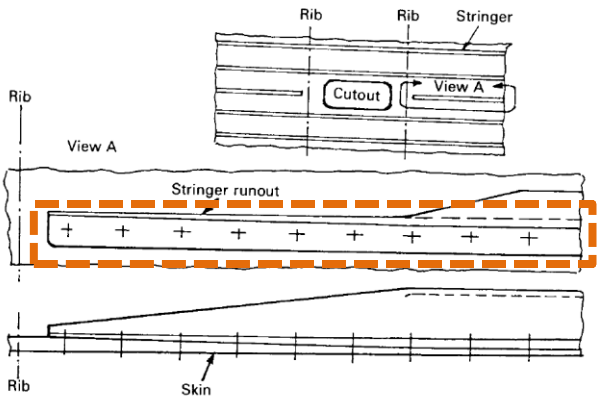
Revision: **1.0**

Date: **07/05/2021**

1 INTRODUCTION

1.1 OBJECT

This chapter defines the methodology to use for obtaining the internal loads in the different elements (rivets or bolts, plates, etc.) involved in a multi-fastener continuous joint (e.g. fastened joints between skin panels, or between a doubler and the stringers ends it is connecting), where the stress problem can be solved in one dimension (quasi-1D analysis; no loads redistributions in other directions).



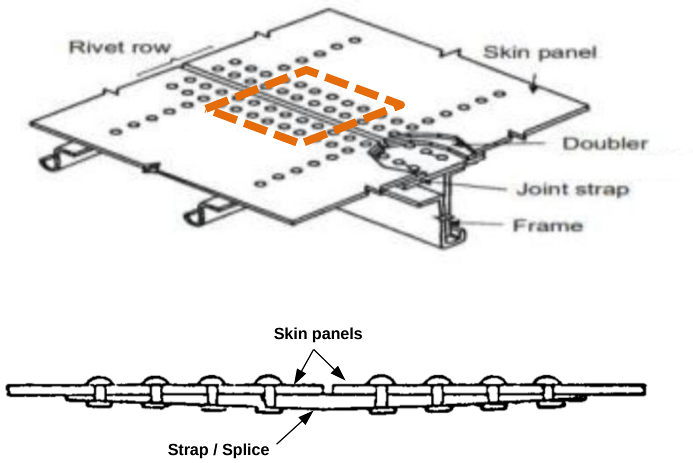


Figure 1. Typical examples of continuous single shear joints.

1.2 SCOPE AND LIMITATIONS

Analysis methodology described in the present chapter is only applicable for the structural joints which have the following characteristics:

- ✓ Single Shear joint. Double Shear joints can be also analysed following this method if the analysis configuration is appropriately transformed to two single shear ones before any calculation.
- ✓ Unidimensional joint. All analysed fasteners are considered to be located in the same row. Joints with more than one row of fasteners can be also analysed following this method if each row is appropriately studied in separated calculations.
- ✓ Loads applied inside the joint main plane, such as tension-lap-shear, in-plane-shear and local thermal loads, are the only ones considered (no out-of-plane loads are covered by method).
- ✓ Joined parts must be sufficiently thin at fastened area in order to avoid any relevant secondary bending effect caused by applied loads.
- ✓ Behaviour of joined parts and fasteners is considered to be linear elastic.
- ✓ Joined parts can be made of both isotropic and/or orthotropic materials.

2 METHOD DESCRIPTION

The non-numerical method proposed herein is based in the classical principles of displacements compatibility between the two joined parts. Each joined part/plate is considered divided in a series of longitudinal elements between fasteners, which are idealized as rods or springs working in tension or compression, through the joint unidimensional main direction, as shown in Figure 2.

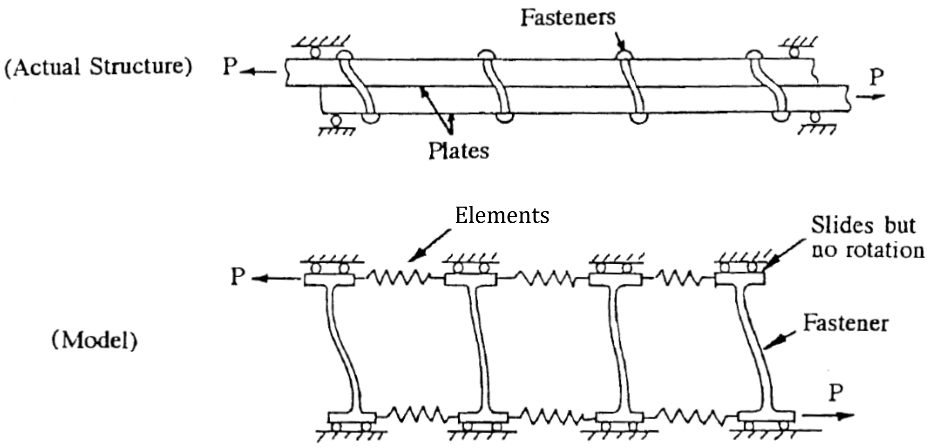


Figure 2. Spring system idealization model of a continuous single shear fastened joint.

Sign criterion considered for axial/shear loads $\boldsymbol{P}$ and $\boldsymbol{Q}$ applied to the continuous joint and their respective reactions is shown in scheme of Figure 3 ($\boldsymbol{L}$: Left end; $\boldsymbol{R}$: Right end), where can be noticed that the effect of axial and shear applied loads must be studied in different analyses.

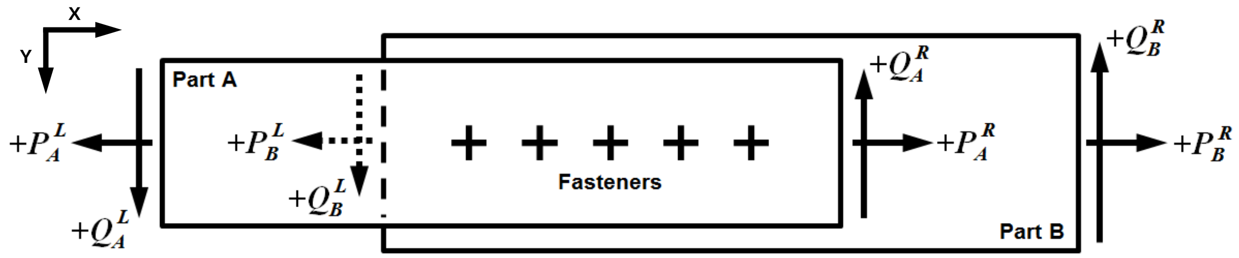


Figure 3. Sign criteria scheme of applied loads/reactions in a continuous single shear fastened joint.

Distribution of internal shear loads reacted by fasteners and internal axial loads at joined parts follow the load paths scheme shown in Figure 4 for n number of fasteners and $n+1$ number of elements.

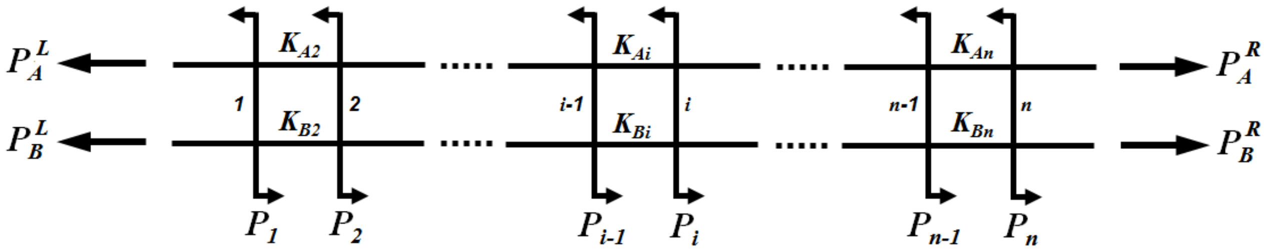


Figure 4. Load paths in a continuous single shear fastened joint.

According to Figure 4, the stiffness values of the elements simulating the joined parts are function of their corresponding modulus, cross-section area and distance between fasteners as follows (see 'Abbreviations and Symbols' in §3 for variables description):

- Tension-lap-shear analyses $\rightarrow$ $K_{A_i} = \frac{E_A \cdot A_{A_i}}{L_i}$; $K_{B_i} = \frac{E_B \cdot A_{B_i}}{L_i}$
- In-plane-shear analyses $\rightarrow$ $K_{A_i} = \frac{G_A \cdot A_{A_i}}{L_i}$; $K_{B_i} = \frac{G_B \cdot A_{B_i}}{L_i}$

  Remark: In case of parts made of an orthotropic material, the modulus value to be considered for elements stiffness calculations must be the one associated to same direction of rivet loads.

Then, the following equations can be defined:

- Relative displacements and local strain of each element i and fastener i ($3n - 2$ equations):
  - Part A elements ($n - 1$ equations) $\rightarrow$ $\delta_{A_i} - \delta_{A_{i-1}} = \frac{F_{A_i}}{K_{A_i}} + \alpha_{A_i} L_i \Delta T$
  - Part B elements ($n - 1$ equations) $\rightarrow$ $\delta_{B_i} - \delta_{B_{i-1}} = \frac{F_{B_i}}{K_{B_i}} + \alpha_{B_i} L_i \Delta T$

- Fasteners (n equations) $\rightarrow$ $\delta_{B_i} - \delta_{A_i} = \frac{P_i}{S_i}$

- Reference displacement (1 equation) $\rightarrow$ $\delta_{A_1} = 0$

- Internal axial loads at joined parts ends –similar for shear loads analysis– (4 equations):
  - Part A ends $\rightarrow$ $F_{A_1} = P_A^L$, $F_{A_{n+1}} = P_A^R$
  - Part B ends $\rightarrow$ $F_{B_1} = P_B^L$, $F_{B_{n+1}} = P_B^R$

- Equilibrium of forces (2n – 1 equations):
  - Part A elements (n – 1 equations) $\rightarrow$ $F_{A_i} = \sum_{j=i}^{n} P_j + P_A^R$
  - Part B elements (n – 1 equations) $\rightarrow$ $F_{B_i} = -\sum_{j=i}^{n} P_j + P_B^R$
  - Fasteners (1 equation) $\rightarrow$ $\sum_{j=1}^{n} P_j = P_A^L - P_A^R$

<u>Remark:</u> In case of in-plane shear analyses, considered elements displacements δ of previous equations must be the ones in transversal Y direction (see axes reference system in Figure 3).

Then, the total number of equations that can be defined for n points is 5n + 2 and total number of unknown variables is also 5n + 2, which are the following ones:

- δ_{A_i} (n unknown variables)
- δ_{B_i} (n unknown variables)
- F_{A_i} (n + 1 unknown variables)
- F_{B_i} (n + 1 unknown variables)
- P_i (n unknown variables)

Previous defined equations can be combined in a unique determined equations system, with n equations and n unknown variables, which must be solved in order to obtain the load in each joint point P_i. According to this, the equations of this system can be ordered as follows:

- First equation $\rightarrow$ $\sum_{j=1}^{n} P_j = P_A^L - P_A^R$

- Rest of n–equations $\rightarrow$ $\left(\frac{1}{S_{i-1}}\right) P_{i-1} - \left(\frac{1}{S_i}\right) P_i - \left(\frac{1}{K_{A_i}} + \frac{1}{K_{B_i}}\right) \sum_{j=i}^{n} P_j = \frac{P_A^R}{K_{A_i}} - \frac{P_B^R}{K_{B_i}} - \left(\alpha_{B_i} - \alpha_{A_i}\right) L_i \Delta T$

Therefore, loads in each joint point P_i can be obtained by solving this linear equations system. Tension-lap-shear and in-plane-shear analyses must be separately solved following this method and, after that, total loads and stresses shall be calculated as combination of the results previously obtained by those separated analyses.

The values of the joint single shear stiffness at each fastener S_i shall be calculated according to the methodology described in §2.1 of Chapter 3210 'Types of Riveted Joints'. Notice that stiffness S_i at fasteners can be assumed to be the same in both tension-lap-shear and in-plane-shear analyses.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
A_{A_i} , A_{B_i}	mm^2	Cross-section area of element i of joined part A or B
α_{A_i} , α_{B_i}	ºC^{-1}	Thermal expansion coefficient of joined part A or B at element i
δ_{A_i} , δ_{B_i}	mm	Displacement in X or Y direction of fastener i at joined part A or B
ΔT	ºC	Temperature increment considered in the analysis
E_A , E_B	MPa	Material Young Modulus of joined part A or B
F_{bry}	MPa	Limit Bearing Stress
F_{A_i} , F_{B_i}	N	Internal load in element i of joined part A or B
i , j	-	Element or Fastener identification indexes, depending on the context
K_{A_i} , K_{B_i}	N/mm	Stiffness of element i of joined part A or B
G_A , G_B	MPa	Material Shear Modulus of joined part A or B
L_i	mm	Length of element i (same for both joined elements)
n	-	Total quantity of fasteners
P_i	N	Load reacted by each fastener i
P_A , P_B	N	External axial loads at joined part A or B
Q_A , Q_B	N	External shear loads at joined part A or B
S_i	N/mm	Joint stiffness in fastener i
TBC	-	To Be Confirmed

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	ESDU 98012	Oct/2001	'Flexibility of, and load distribution in, multi-bolt lap joints subject to in-plane axial loads'.
Ref. 2	NT-T-ADP-99026	A	'Theoretical Manual of Program ARES v.2.0' [Original in Spanish].

SOFTWARE

Program	***Version***	***Description***
-	-	-

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
1.0 19/03/2020	Guillermo Moreda	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Methods (TEPAP-TL4)	G. Baños 07/05/2021	F. Glock 07/05/2021

Table 3. *Approval signatures table for last revision of this chapter*